

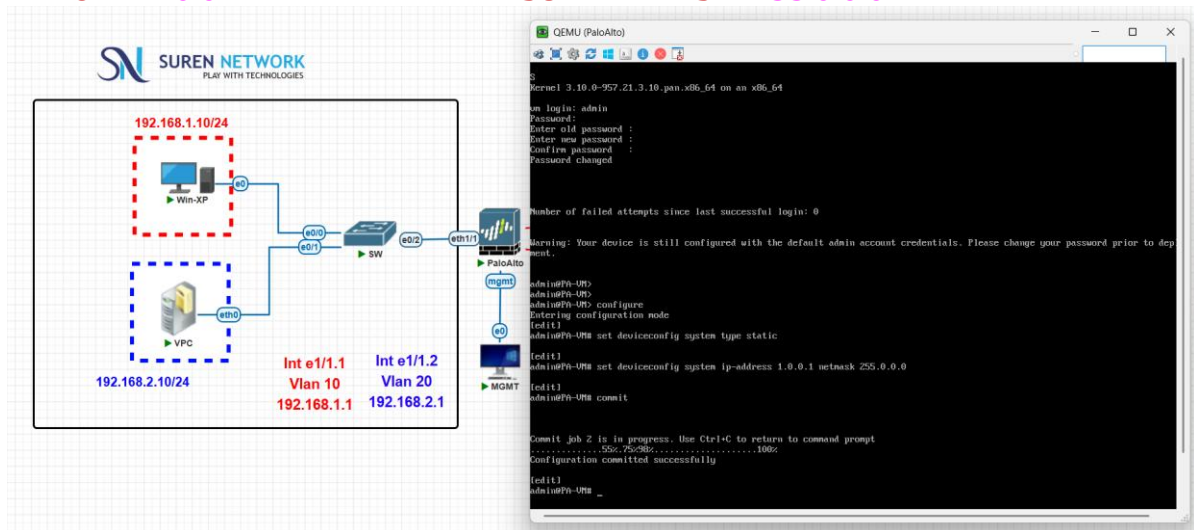
LAB 1 ECMP INTERVLAN

GURUCHANDHRAN M

TASK 1: FIREWALL IP ASSIGNMENT

PALO IP: 1.0.0.1

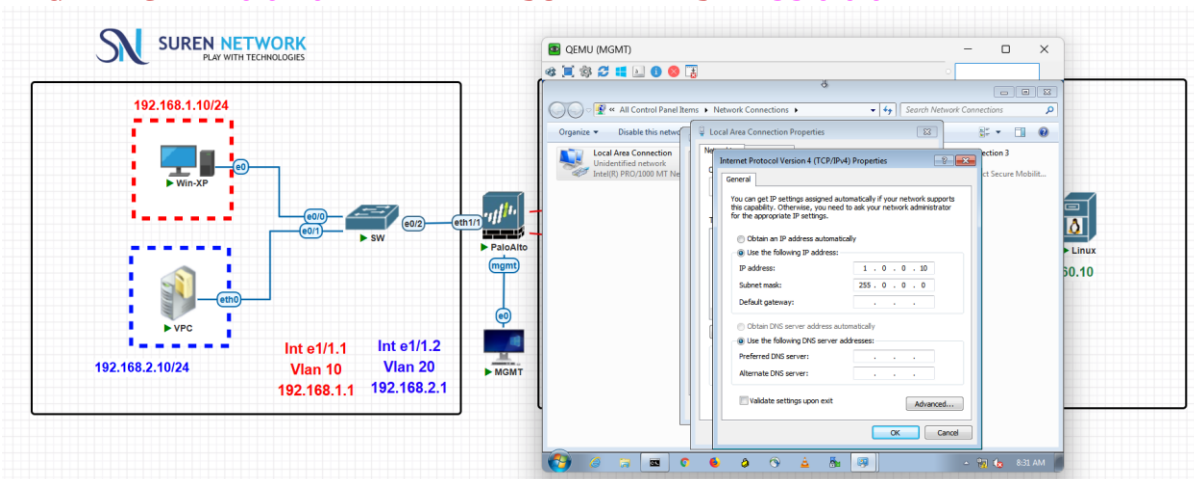
SUBNET MASK: 255.0.0.0



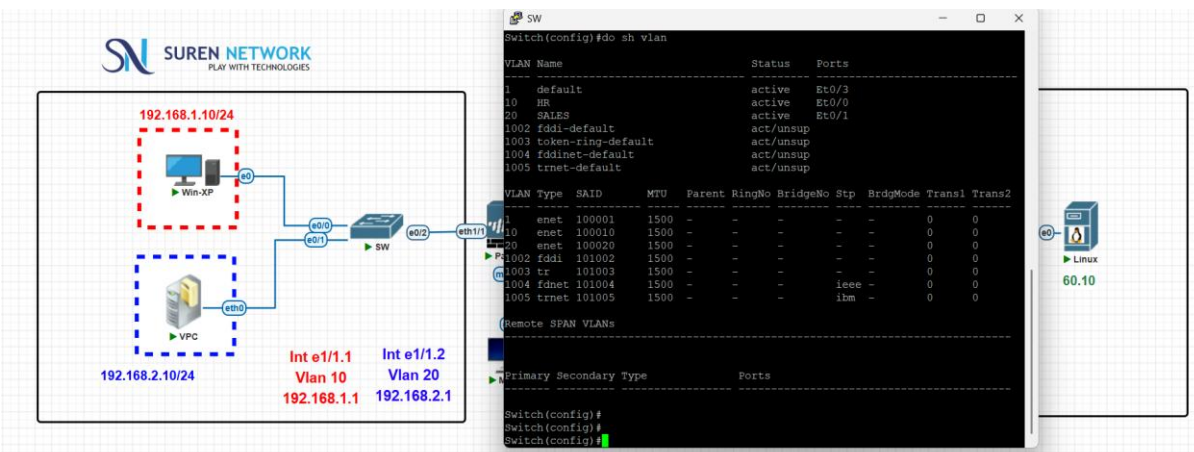
TASK 2: MANAGEMENT PC IP ASSIGNMENT

MGMT PC IP: 1.0.0.10

SUBNET MASK: 255.0.0.0



TASK 3: VLAN CONFIGURATION IN SWITCH



TASK 4: CONFIGURING LAN ZONES IN THE NETWORK

QEMU (MGMT) PA-VM

https://1.0.0.1/#network:vsys1:network/zones

Zones

NAME	TYPE	INTERFACES / VIRTUAL SYSTEMS	ZONE PROTECTION PROFILE	PACKET BUFFER PROTECTION	LOG SETTING	ENABLED	INCLUDED NETWORKS	EXCLUDED NETWORKS	ENABLED	INCLUDED NETWORKS	EXCLUDED NETWORKS
HR	layer3			☒		☐	any	none	☐	any	none
SALES	layer3			☒		☐	any	none	☐	any	none

TASK 5: SETTING UP INTERFACES (IP Assignments, Default Virtual Routers)

QEMU (MGMT) PA-VM

https://1.0.0.1/#network:vsys1:network/interfaces

Interfaces

INTERFACE	INTERFACE TYPE	MANAGEMENT PROFILE	LINK STATE	IP ADDRESS	VIRTUAL ROUTER	TAG	VLAN / VIRTUAL-WIRE	SECURITY ZONE
ethernet1/1	Layer3		☒	none	none		Untagged	none
ethernet1/1.1	Layer3		☒	192.168.1.1/24	default	10	none	HR
ethernet1/1.2	Layer3		☒	192.168.2.1/24	default	20	none	SALES
ethernet1/2			☒	none	none		Untagged	none
ethernet1/3			☒	none	none		Untagged	none

TASK 6: CREATING NETWORK OBJECT (HR-LAN, SALES-LAN) FOR LAN NETWORK

QEMU (MGMT) PA-VM

https://1.0.0.1/#objects:vsys1:objects/addresses

Addresses

NAME	LOCATION	TYPE	ADDRESS	TAGS
HR-LAN		IP Netmask	192.168.1.0/24	
SALES-LAN		IP Netmask	192.168.2.0/24	

TASK 7: DRAFTING SECURITY POLICIES FOR LAN TO REACH INTERNET

The network diagram shows a Win-XP VM connected to a VPC via a bridge. The Win-XP VM has IP 192.168.1.10/24. The VPC has IP 192.168.2.10/24. The bridge is connected to the Internet (eth0/1). The VPC is connected to the Internet via a Vlan 10 interface.

The QEMU (MGMT) interface shows the PA-VM configuration. The Security tab is active, displaying a table of security policies:

NAME	TAGS	TYPE	ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS	DEVICE
1 COMMUNICATION	none	universal	HR	HR-LAN	any	any	HR	HR-LAN	any
2 intrazone-default	none	intrazone	any	any	any	any	(intrazone)	any	any
3 interzone-default	none	interzone	any	any	any	any	any	any	any

TASK 8: PC IP ASSIGNMENT

PC1 IP: 192.168.1.10

PC2 IP: 192.168.2.10

The first screenshot shows the QEMU (Win-XP) interface with the Network Connections window open. The Internet Protocol (TCP/IP) Properties window is displayed, showing the IP address 192.168.1.10 and subnet mask 255.255.255.0.

The second screenshot shows the VPC interface with the command prompt open. The command `ip add 192.168.2.10 255.255.255.0` is entered, and the output shows the IP address assigned successfully.

TASK 9: CONNECTION ESTABLISHMENT VERIFICATION USING ping

The first screenshot shows the QEMU (Win-XP) interface with the command prompt open. The command `ping 192.168.2.10` is entered, and the output shows the ping successful.

The second screenshot shows the VPC interface with the command prompt open. The command `show ip` is entered, and the output shows the IP configuration for VPC[1]:

```
VPC> show ip
NAME       : VPC[1]
IP/MASK    : 192.168.2.10/24
GATEWAY    : 192.168.2.1
DNS        :
MAC        : 00:50:79:66:68:6a
LPORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPC> ip 192.168.2.10 255.255.255.0 192.168.2.1
Checking for duplicate address...
VPC : 192.168.2.10 255.255.255.0 gateway 192.168.2.1

VPC> ping 192.168.1.10
84 bytes from 192.168.1.10 icmp_seq=1 ttl=127 time=4.116 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=127 time=3.163 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=127 time=3.952 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=127 time=4.356 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=127 time=3.497 ms
VPC>
```

TASK 10: CONFIGURING WAN ZONES IN THE NETWORK

NAME	TYPE	INTERFACES / VIRTUAL SYSTEMS	ZONE PROTECTION PROFILE	PACKET BUFFER PROTECTION	LOG SETTING	User-ID	Device-ID
HR	layer3	ethernet1/11		☒		any	any
SALES	layer3	ethernet1/12		☒		any	any
WAN 1	layer3			☒		any	any
WAN 2	layer3			☒		any	any

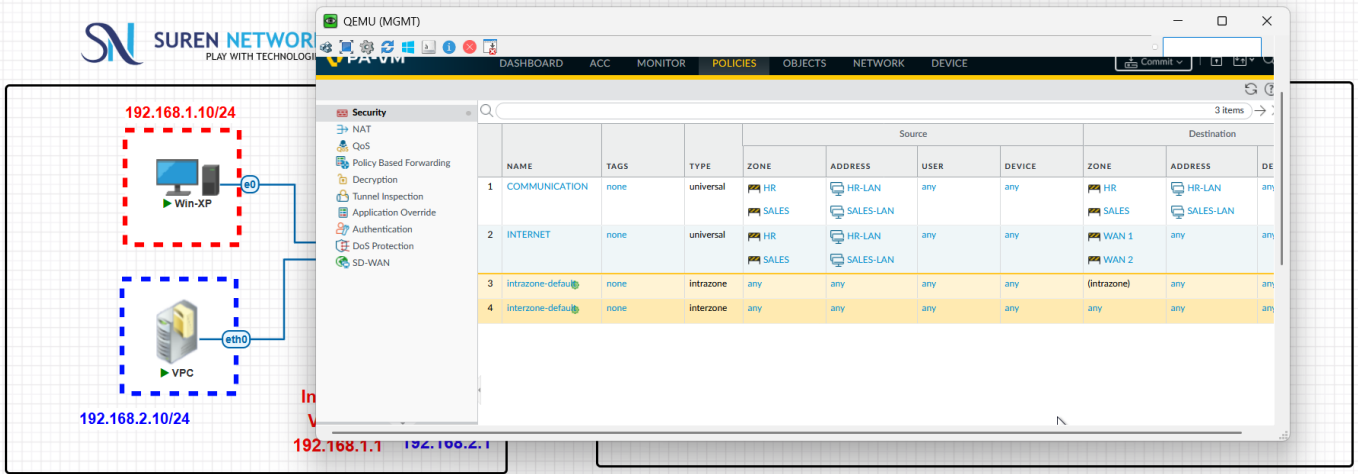
TASK 12: SETTING UP WAN INTERFACES (IP Assignments, Default Virtual Routers)

INTERFACE	INTERFACE TYPE	MANAGEMENT PROFILE	LINK STATE	IP ADDRESS	VIRTUAL ROUTER	TAG	VLAN / VIRTUAL-WIRE	SECURITY ZONE
ethernet1/1	Layer3		☒	none	none	Untagged	none	none
ethernet1/1.1	Layer3		☒	192.168.1.1/24	default	10	none	HR
ethernet1/1.2	Layer3		☒	192.168.2.1/24	default	20	none	SALES
ethernet1/2	Layer3		☒	20.0.0.1/8	default	Untagged	none	WAN 1
ethernet1/3	Layer3		☒	30.0.0.1/8	default	Untagged	none	WAN 2
ethernet1/4	Layer3		☒	none	none	Untagged	none	none

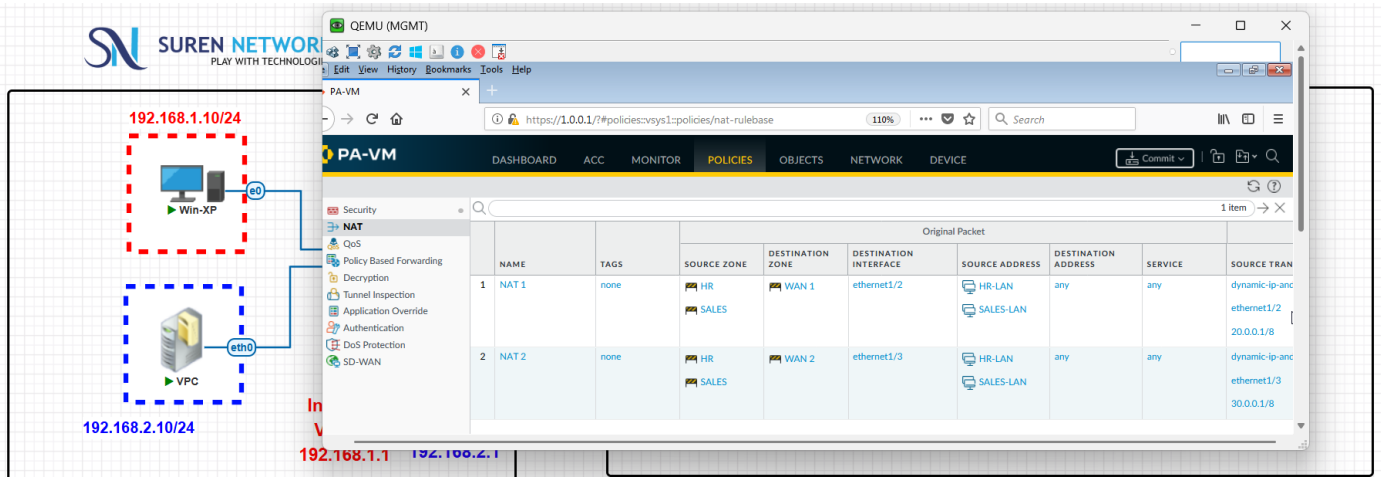
TASK 13: ASSIGNING DEFAULT ROUTES TO ISP 1 AND ISP 2

NAME	DESTINA...	INTERFA...	TYPE	VALUE	ADMIN DISTANCE	METRIC	BFD	ROUTE TABLE
ISP-1	0.0.0.0/0	ethernet1...	ip-address	20.0.0.2	default	10	None	unicast
ISP-2	0.0.0.0/0	ethernet1...	ip-address	30.0.0.2	default	20	None	unicast

TASK 14: DRAFTING SECURITY POLICIES FOR WAN TO REACH INTERNET



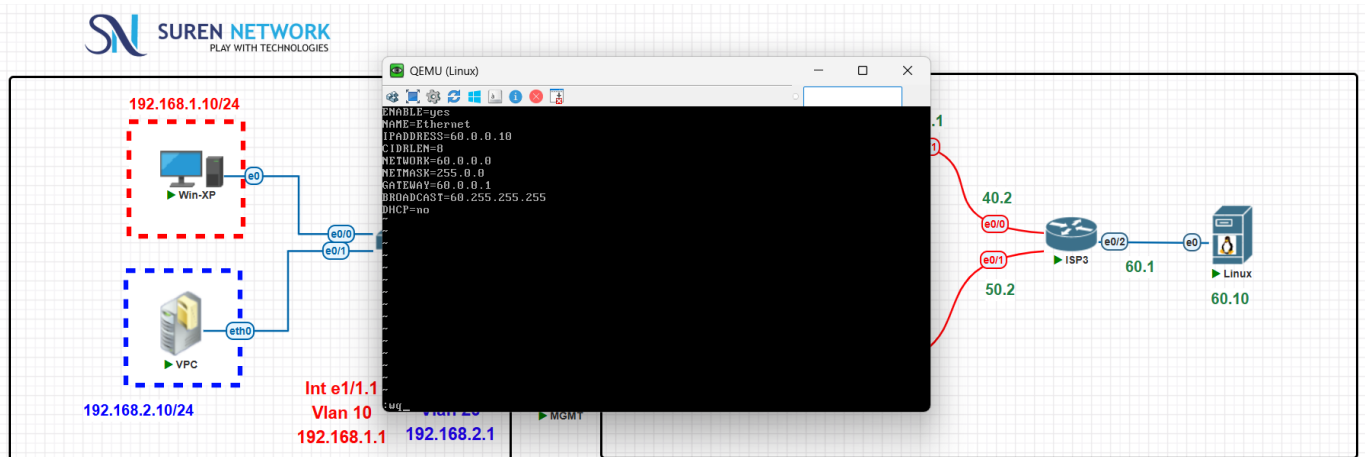
TASK 15: ENABLING NAT POLICIES FOR WAN COMMUNICATION



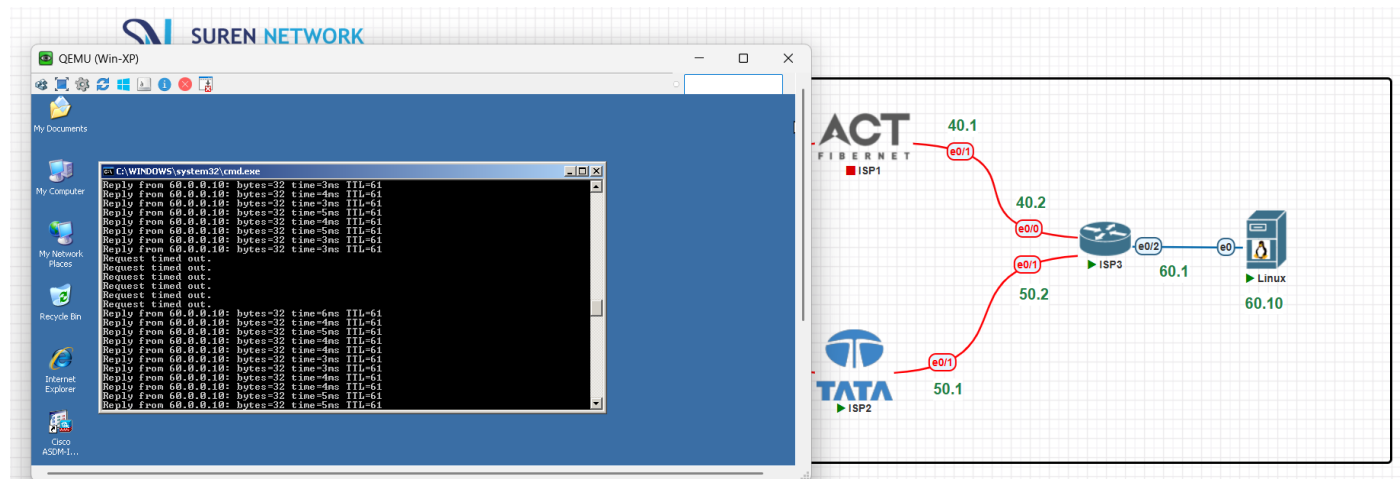
TASK 16: ASSIGNING IP TO LINUX SERVER

SERVER IP: 60.0.0.10

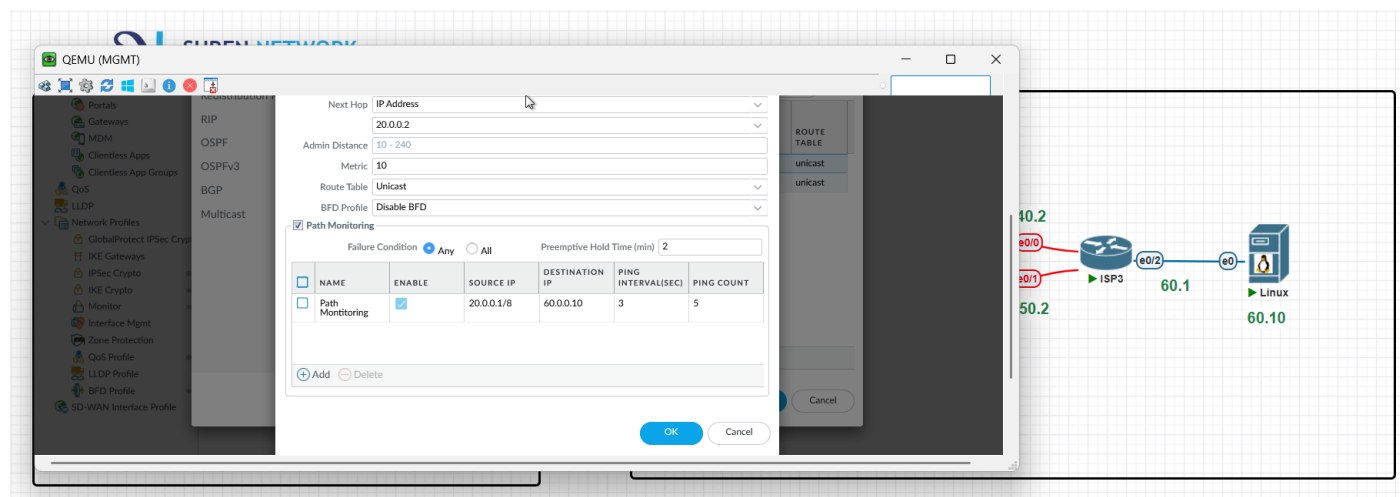
SUBNET MASK: 255.0.0.0



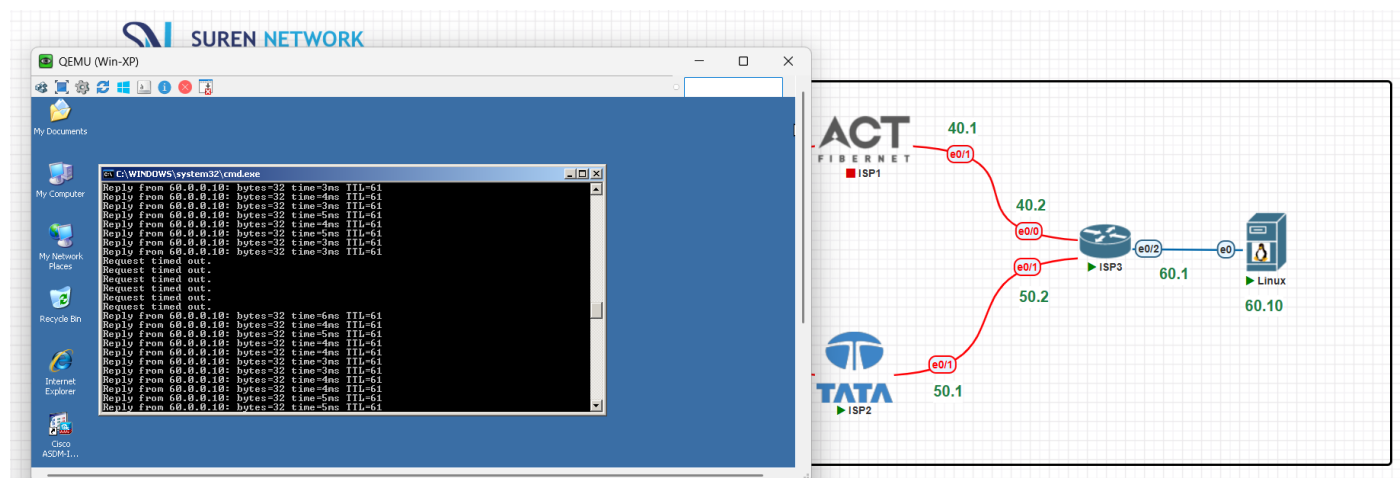
TASK 17: CONNECTION ESTABLISHMENT VERIFICATION USING ping



TASK 18: ENABLING PATH MONITORING FOR WAN INTERFACE



TASK 19: CONNECTION ESTABLISHMENT VERIFICATION USING ping



TASK 20: ADDING POLICY BASED FORWARDING POLICIES

The screenshot shows the QEMU (MGMT) Policy Based Forwarding configuration window. The table below lists the configured policies:

	NAME	TAGS	ZONE/INTERFACE	ADDRESS	USER	ADDRESS	APPLICATION	SERVICE	ACTION
1	HTTP	none	HR	HR-LAN	any	60.0.0.10	any	service-http	forward
2	SSH	none	HR	HR-LAN	any	60.0.0.10	any	ssh	forward

The network diagram shows a router (ISP3) connected to a Linux host (60.10). The router has interfaces e0/0, e0/1, and e0/2. The Linux host has interface e0. The router is connected to the Linux host via e0/2 and e0. The router is also connected to the Internet via e0/0 and e0/1.

TASK 21: VERIFYING HTTP FROM WEB BROWSER USING POLICY BASED FORWARDING POLICIES

The screenshot shows the QEMU (Win-XP) web browser displaying a successful HTTP connection to the URL `http://60.0.0.10/`. The browser shows the text "Hello World! This file is located in /home/httpd".

The network diagram shows the same setup as Task 20, but with an additional ISP1 (40.1) connected to the router (ISP3) via interface e0/1. The router is connected to the Linux host (60.10) via interface e0/2. The router is also connected to the Internet via interface e0/0.

TASK 22: VERIFYING SSH FROM PUTTY USING POLICY BASED FORWARDING POLICIES

The screenshot shows the QEMU (Win-XP) Putty terminal window. The terminal displays the output of the `ifconfig` command, showing the network configuration for the Linux host (60.0.0.10). The output is as follows:

```
root@60.0.0.10's password:
Chop wood, carry water.
#
# ifconfig
eth0      Link encap:Ethernet  HWaddr 00:50:00:00:08:00
          inet addr:60.0.0.10  Bcast:60.255.255.255  Mask:255.0.0.0
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:341 errors:15 dropped:0 overruns:0 frame:15
          TX packets:197 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:33986 (33.1 KiB)  TX bytes:16211 (15.8 KiB)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:0 errors:0 dropped:0 overruns:0 frame:0
          TX packets:0 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:0 (0.0 B)  TX bytes:0 (0.0 B)
```

The network diagram shows the same setup as Task 20, but with an additional ISP1 (40.1) connected to the router (ISP3) via interface e0/1. The router is connected to the Linux host (60.10) via interface e0/2. The router is also connected to the Internet via interface e0/0.